Speaker Role Identification in Clinical Conversations

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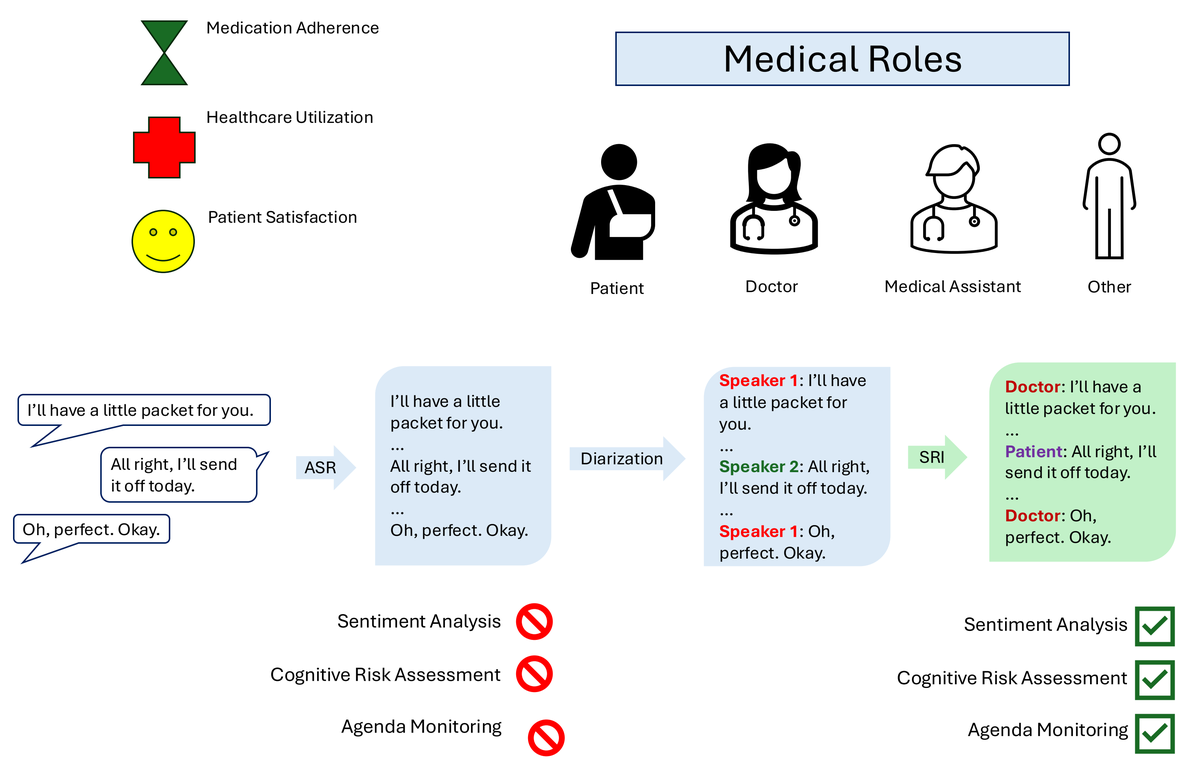
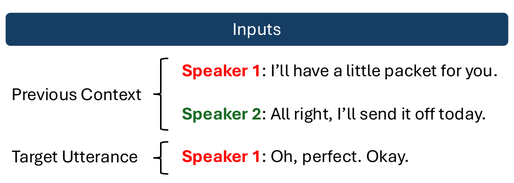
Introduction

- Patient-clinician communication drives functional outcomes[1,2,3] and merits further study[4].
- Studying clinical communication involves analyzing conversational transcripts.
- Studying clinical transcripts effectively requires determination of who is speaking at each turn.
- Modern Automated Speech Recognition (ASR) tools can automatically transcribe conversations into turns, and "speaker diarization" software can assign numeric speaker identifiers at each turn based on voice.
- However, these tools do not assign speaker roles to turns. **The task of Speaker Role Identification (SRI) instead tends to be manual.**
- In non-clinical domains, LLMs have been used to infer roles based on text[5,7,8] and numeric speaker identifiers[5,6].
- However, ***nobody has investigated SRI in real-world clinical settings***.
- We explore whether Large Language Models (LLMs) such as BERT can accurately perform SRI in clinical transcripts, with and without diarization

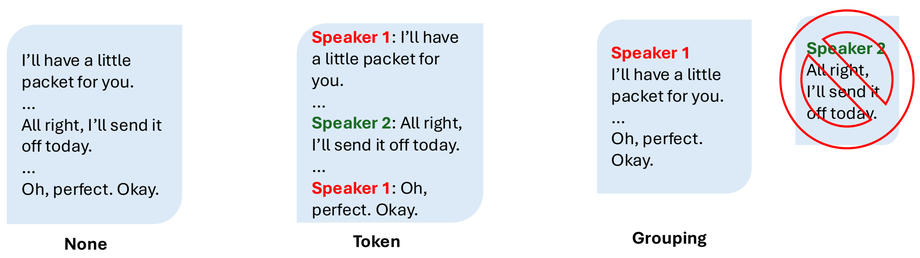


Methods

- **Dataset:** 117 manually transcribed clinical visit transcripts from the EF Study[9] (27,505 labeled utterances)
- **Task:** Tag individual utterances with speaker role using LLMs.
- **Model**: BERT base-cased
- **Inputs:** Target utterance + context (variable size), speaker identifiers



- **Use of Diarization:**
  - None vs. grouping method vs. token method
  - Simulated diarization identifier errors as ablation study



- **Baseline/Comparison Models:**
  - (Boosted) Decision Trees
    - Bag of words variant
    - Heuristics variant (pronoun usage + time within visit statistics)

Results

Model Performance

- BERT with no diarization: 82% F1/Acc; with accurate diarization identifiers: 95% F1/Acc
- LLMs outperform decision trees, and text inputs outperform heuristics
- Expanding context size improves metrics up to a limit

Diarization and Context Impact

- Diarization consistently improves all model types
- Token strategy outperformed grouping strategy

				Diarization				No Diarization
Model	Diarization	Inputs	Context Size	50% error	25% error	10% error	0% error	
Decision Tree	N/A	Heuristics	1	N/A	N/A	N/A	N/A	0.62/0.58
Decision Tree	Grouped	Heuristics	2	0.58/0.54	0.60/0.56	0.61/0.58	0.62/0.59	N/A
Decision Tree	Grouped	Heuristics	5	0.62/0.61	0.67/0.66	0.70/0.70	0.71/0.71	N/A
Decision Tree	Grouped	Heuristics	10	0.64/0.63	0.70/0.69	0.74/0.73	**0.76/0.76**	N/A
Decision Tree	N/A	Bag of Words	1	N/A	N/A	N/A	N/A	0.62/0.57
Decision Tree	Grouped	Bag of Words	2	0.55/0.51	0.58/0.54	0.60/0.55	0.62/0.56	N/A
Decision Tree	Grouped	Bag of Words	5	0.58/0.56	0.62/0.60	0.67/0.64	0.71/0.68	N/A
Decision Tree	Grouped	Bag of Words	10	0.55/0.53	0.62/0.59	0.68/0.66	**0.73/0.71**	N/A
Boosted Trees	N/A	Heuristics	1	N/A	N/A	N/A	N/A	0.62/0.59
Boosted Trees	Grouped	Heuristics	2	0.58/0.55	0.60/0.57	0.61/0.58	0.62/0.59	N/A
Boosted Trees	Grouped	Heuristics	5	0.63/0.62	0.67/0.66	0.71/0.70	0.72/0.71	N/A
Boosted Trees	Grouped	Heuristics	10	0.65/0.64	0.71/0.70	0.74/0.74	**0.77/0.77**	N/A
Boosted Trees	N/A	Bag of Words	1	N/A	N/A	N/A	N/A	0.65/0.61
Boosted Trees	Grouped	Bag of Words	2	0.56/0.52	0.60/0.56	0.63/0.59	0.65/0.61	N/A
Boosted Trees	Grouped	Bag of Words	5	0.63/0.61	0.67/0.64	0.72/0.70	0.76/0.74	N/A
Boosted Trees	Grouped	Bag of Words	10	0.58/0.55	0.67/0.65	0.76/0.74	**0.82/0.80**	N/A
BERT	N/A	Sequence	1	N/A	N/A	N/A	N/A	0.73/0.73
BERT	Token	Sequence	2	0.86/0.86	0.90/0.90	0.90/0.90	**0.95/0.95**	0.82/0.82
BERT	Token	Sequence	5	0.86/0.86	0.89/0.89	0.92/0.92	**0.95/0.95**	0.80/0.80
BERT	Token	Sequence	10	0.71/0.71	0.75/0.76	0.81/0.81	0.84/0.84	0.68/0.68
BERT	N/A	Sequence	1	N/A	N/A	N/A	N/A	N/A
BERT	Grouped	Sequence	2	0.69/0.67	0.69/0.67	0.68/0.66	0.70/0.68	N/A
BERT	Grouped	Sequence	5	0.71/0.70	0.74/0.74	0.79/0.78	0.83/0.83	N/A
BERT	Grouped	Sequence	10	0.70/0.69	0.76/0.76	0.82/0.82	**0.88/0.88**	N/A

Conclusion

- Semantic features enable robust speaker role identification in clinical transcripts, with or without diarization.
- Diarization enhances accuracy when well implemented and is most effective when incorporated flexibly.
- SRI offers new opportunities for downstream analysis, including role-based segmentation and clinical quality measurement at scale.

LLMs can accurately perform SRI in clinical conversation transcripts.

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